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TOWARD SECURE DOOR LOCK SYSTEM: DEVELOPMENT IOT SMART DOOR LOCK DEVICE

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A B S T R A C T

Conventional doors have gotten little attention for a number of reasons, including the need for a unique key for each door, the requirement of specialized equipment to duplicate keys, the difficulty to determine who opened the door, and the absence of alternatives to utilizing the traditional key. Apart from that, standard locks are subject to lock picking tools, tools that reach under/over the door frame to reach the door handle, and bump keys, which utilize force to set the lock pins and open the door. This paper develops a secure Internet of Things Smart Door Lock System (IoT-SDL) that can unlock door without keys or other physical tools. Installing and initializing the IoT-SDL system requires inserting the home's WIFI SSID and password into the Raspberry Pi's WIFI access point and providing internet connectivity. After installation, users may be configured and enabled IoT-SDL capabilities using a web interface. The IoT-SDL system includes a touchscreen keypad and an RFID scanner that are used to unlock the door if the user chooses to activate the two-factor authentication function. The results showed the ability of the IoT-SDL system to remember who pushed the door unlock button, so it is possible to exactly know who has been in the space. Additionally, the user had the ability to generate temporary PIN codes that are valid for a predetermined amount of time and are intended for reliable guests to enter on the touchscreen keypad of the IoT-SDL in order to open the door. The IoT-SDL system includes a number of additional options that can be used to personalize the way the door functions due to the essential property that the door should be used to serve its users rather than the other way around. The user had the ability to remotely open the door with the push of a button after they have used the interfaces that are provided to them. The access was possible to any and all of the data that the IoT-SDL has collected by making use of the different application program interfaces.

1. INTRODUCTION

The home automation system is the most rapidly developing new technology. It consists of a number of intelligent home equipment designed to make life easier for homeowners. Included in a home automation system is home security in the form of access control systems, appliance management, temperature control, and lighting control. Embedded systems, smart sensors, and wireless technology together with recent advancements

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in interactive applications have all contributed to an improvement in the quality of human existence in a variety of sectors like home automation system, smart healthcare, and smart automobiles. Several different kinds of household appliances, such as refrigerators, sewing machines, washing machines, televisions, and so on, are now equipped with built-in sensors that enable them to function as part of a networked smart home system. As a result, there is a pressing need to pay careful attention to the protection of both life and property, which is especially true now that the age of 5G and the internet of things (IoT) has arrived. Security in the home is really necessary. The growing number of criminal activities throughout the globe is a significant factor in the importance of home security. The IoT-Smart Door Lock (IoT-SDL) system provides a barrier for illegal activities. IoT-SDLs are going to be everyone's go-to choose for their home improvement projects. They provide the same degree of security as conventional door locks while also providing a variety of replacements for the conventional key. In the past, there was only one type of authentication that could be used for door locks, and that was the use of a mechanical key. On the other hand, the keys are often lost, forgotten at home, or misplaced. IoT-SDLs provide users with complete assurance by providing a range of authentication methods from which they may choose one that is suited to their individual requirements (Wu et al., 2023).

The rest of paper is organized as follows. In section 2, a deeply research considering the use of the system mentioned in the Literature Review, the IoT-SDL model and its functional and non-functional requirements are presented in the Methodology section 3. In Section 4, the IoT-SDL system diagram in term of class, sequence, and Entity Relation (ER) are introduced. In section 5, the code of the IoT-SDL is presented and tests are implemented to check the performance of the IoT-SDL system code. In section 6, the results of the developed IoT-SDL system are introduced and discussed. The conclusions and future work are presented in section 7.

2. LITERATURE REVIEW

These days, numerous homes and apartments are available for rent to tourists via websites such as Airbnb. For the sake of convenience for both the visitor and the host, there should be a method to hand over a temporary access key to the guest. In (Pardeshi et al., 2023), an electronic door lock mechanism that is capable of producing dynamic temporary access keys for guests is presented. The smartphone serves as the controller for the door lock, and the dynamic key is sent to the smartphone through a Bluetooth Low Energy connection. The generation of block cipher keys is accomplished via the use of fusion techniques (artificial neural networks) and block cipher encryption in the process of developing dynamic keys. The generator generates a one-of-a-kind dynamic key that is valid for a certain amount of time, and the nRF8001 demonstrates that it is a trustworthy Bluetooth Low Energy transmitter and receiver. As a component of the IoT, smart door locks have recently seen widespread use IoT. Yet, reports in the media have indicated that digital door locks have been bypassed by unauthorized users in order to gain access to private residences and commercial buildings. There are variety of security issues that are presented in IoT systems. In the event that these issues are not solved as soon as possible, the growing number of linked devices will make them impossible to monitor. When designing an IoT system, security issues need to be included right from the start. The second reason is that IoT devices are often restricted devices, which means they do not have the same capabilities as regular computers. As a result, the majority of IoT devices do not have the capability to carry out contemporary secure communication protocols. Hardware implementation of cryptographic algorithms and secure communication protocols are features seen in modern electronics (Motwani et al. 2023). A digital door lock system that is compatible with the IoT environment is suggested to be implemented in (Rajput et al., 2024). It is planned and executed with the purpose of enhancing both security and convenience. When an unauthorized user makes an attempt to perform an illegal operation, the proposed system enhances the existing security functions by sending alarm information and recorded images to the user's mobile device. Additionally, the system can send alarm information to the user's mobile device if the door lock sustains any kind of physical damage. The user's ability to examine the access information and remotely activate the door lock is one of the many convenient features that the suggested system offers. As a result of the explosion of the IoT, several creative approaches have been developed to handle the issues that have arisen in the sector of security. A poll conducted in the United States found that about one quarter of Americans had a tendency to forget their home keys twice a week. Furthermore, more than fifty percent of respondents said that the missing things routinely cause them to be late to work or school. With these considerations, a smart lock system that is simple to use is developed in (Obruche et al., 2024). The purpose of this system was to present a smart lock system that can make use of a GSM module to accept calls, check the caller ID that is received with a list of registered numbers, and inform the user in the event that there has been illegal access. The operation of locking or unlocking the door is carried out according to the state of the lock at the time, which may be either LOCKED or UNLOCKED, depending on whether or not the caller ID shows an authorized user. In (Tribuana et al., 2024), a prototype is developed for a smart door that included a facial recognition capability. The door may be opened without the homeowner having to make direct touch with the lock by using a facial recognition system that is based on a camera. Since the coronavirus disease 2019 (COVID19) pandemic is so widespread, it is essential to avoid any kind of physical contact. The Raspberry Pi 3B was employed as the primary controller, and a servo motor was put to use in order to operate the locking door actuator. Platforms such as Node-RED, Blynk, and message queue telemetry transport (MQTT), which are particularly strong tools for constructing IoT devices, were used in the development of the software. Python was used to write the code for each and every one of the apps. At the stage devoted to face recognition, the Haar cascade and local binary pattern histogram approaches were put into action. Dialogflow and Firebase are examples of Google Cloud services that were used in the process of integrating Google Assistant. The facial recognition technology and the intelligent door were successfully integrated. If a face was detected, the smart door would unlock (the average threshold for recognition is 60%). An email message with a face picture is sent to the owner of the home if a face is not successfully detected by the camera. The Google Assistant has a success percentage of 92.8% based on 147 different experiments when it came to properly handling customer requests. The design of a Smart Door Lock System with Automation and Security is introduced in (Sayeduzzaman et al., 2024), which is an application of the IoT. The consumers were able to monitor and operate their houses by utilizing an Android mobile, which is made possible by the system. This system was primarily intended for use with residential door locks; however, the design may be modified to accommodate a wide variety of locks in response to specific requirements. It makes use of a Raspberry pi as its primary device, and Raspbian is installed in the vicinity of the lock as its operating system. A variety of sensors have been networked together in order to recognize the doorbell and the knock. Also, it alerts if it detects somebody acting suspiciously. Also, the device was able to monitor if the door is open or closed and record this information. Together with a photo, it sends a notification to the user if there is a visitation or a suspicious individual that has been identified by the mobile application. The system has resorted to using the Google Cloud Messaging service for its notification needs. In the event that the user has reason to be suspicious, they have the ability to place an emergency call as well as open the door for a recognized guest. After the verification of the pin code was complete, the door will be opened. It would be possible for a single application to control and monitor several locks simultaneously. In (Kaur et al., 2023), how secure home smart locks are taken in consideration, which are cyber-physical devices that replace conventional door locks with deadbolts that can be electronically controlled by mobile devices or the distant servers of the lock maker. It discussed two different types of assaults that may be carried out on smart locks, and then evaluate the level of protection provided by five different locks that are currently on the market. The introduced security analysis has shown that there are flaws in the design, implementation, and interaction models of existing locks that can be exploited by a variety of different types of adversaries. This gives these adversaries the ability to learn private information about users and gain unauthorized access to their homes. It is shown a number of potential protections that may be implemented in smart locks and other IoT devices to prevent the vulnerabilities that are highlighted in the introduced demonstrations. One of these protections is a unique method that are currently prototyped and evaluated, and it is designed to securely and usably convey the intended actions of a user to smart locks. In (Guntur J et al., 2023), a novel approach that deals with a security-based system that was conceived with the notion to provide a smart door lock system to replace the conventional lock and key module is presented. Additionally, the purpose of this approach was to protect residences from burglaries when we are not there to protect them. Over the course of the past few years, there has been a shift in the method by which a door lock is operated. This shift has been accompanied by a great deal of innovation thanks to the implementation of technologies such as RFID; however, this has not resolved the issues that are associated with portable objects such as keys, cards, and so on. An electronic door lock that has a gear motor to assist the mechanical system of locking and unlocking the door has had a total of four modules integrated into it as part of the implementation process. For the purpose of controlling the whole system, an Arduino MEGA microcontroller has been implemented. The four modules are as follows: the fingerprint module, which employs the fingerprint sensor; the message module, which makes use of GSM; the passcode module, which makes use of a keypad; and the voice controller module, which makes use of a Bluetooth module. In order to develop and illustrate the suggested method, a three-dimensional model of a home has been employed. The IoT technology and the application of smartphone communication technology to traditional equipment (door lock) are used in the suggested technique in (Goyal PK et al., 2023) to enable remote opening and closing of a door via authentication. In specifically, it suggested a security upgrade plan that is based on the smart door lock system as a solution to the problem with safety that is created by the physical key that is utilized in unmanned automation equipment such as ATMs, KIOSKs, and vending machines. In (Sayeduzzaman et al., 2024, Mohankumar et al., 2024), a ZigBee module is installed inside of a digital door lock, and the door lock itself serves as the central core controller of the whole home automation system. the suggested system may be conceptualized as a distributed network of sensor nodes and actuators, with a digital door lock serving as the base station. The RFID reader for user authentication, touch LCD, motor module for opening and closing the door, sensor modules for detecting the condition inside the house, communication module, and control module for controlling other modules are the components that make up the door lock system that is proposed. Sensor nodes for environment sensing are put at suitable areas at house. The digital door lock serves as the centralized controller, and it is capable of monitoring and controlling the status of each individual ZigBee module. Because the door lock is the very first and very last thing that people encounter when entering and leaving the home, respectively, the home automation function that was included in digital door lock systems makes it possible for users to control and monitor the environment and conditions of their homes in a streamlined manner all at once before entering or leaving the home. In addition to this, it gives consumers the ability to remotely monitor the temperature and humidity levels within the home by using the internet or any other kind of public network. The fact that the suggested system can be readily implemented whenever and wherever it is required without the necessity of any infrastructures or appropriate planning is the primary benefit that the proposed system had. In (Hossain et al., 2024), Prathapagiri and Eethamakula started by discussing the necessity of home security and how IOT devices play a part in it. Then, they described existing market products that employ numerous means of validating a homeowner's identification, such as RFID, Fingerprint, Face ID, PIN codes, and Passwords. They suggested a system that utilizes an ESP32 with a Camera module and has face recognition capabilities, but they don't explain how or if they built their facial recognition algorithm. The system they created seems to simply utilize the ESP32's Bluetooth Low Energy (BLE) capabilities to identify the homeowner's phone and the ESP32's camera module to capture photographs. The door locking mechanism they employed is a solenoid lock. They utilized an FTDI232 USB to SERIAL Interface Board to connect to their device for data transfer, and they leveraged the ESP32's WIFI capabilities to interact wirelessly with the Blynk program. Using the Blynk app as a smartphone interface to the device seems to provide a basic control over the device, such as remotely unlocking and capturing pictures using the ESP32's camera. In (Alkhazali et al., 2023), Husni et al. begin their study by detailing the global rise of IOT devices, and then especially in their home nation of Indonesia. After that, they discussed the components utilized in their device, such as the Raspberry Pi, camera, microswitch (DC motor), and Arduino GSM. Their device logs in using an Android smartphone to the online interface of the smart door system, which is constructed using the PHP programming language on the Laravel Framework, or the Android app produced using the Java programming language. The gadget identifies the user's coordinates over GPRS network utilizing the Arduino GSM. In (Alkhazali et al., 2023), (Vardakis et al., 2024), several components are included in the project by discussing the system and how to integrate the application with the Arduino gadget. In general, an automated door unlocking system is one of the home automation strategies. It is also known as a Bluetooth key, since the user's cell phone serves as the key to open the door. The door may be opened using the phone's Wi-Fi, which includes a camera that sends a picture of the person(s) present at the entrance to the smartphone. It also has a small PIC microcontroller with a 5-digit password for locking and unlocking the doors using a geared motor. If the efforts fail more than three times, it delivers a warning message. The security mechanism for the door lock was created utilizing a graphical user interface (GUI) with facial recognition and a MATLAB PIC microcontroller. In (Goalit et al., 2020), many components are included in the SDL system and how they make their system more secure by detailing the system and how can integrate the application with the Raspberry Pi device. With multiple input models, such as a power supply and a Raspberry Pi3 camera, the system becomes more efficient. Moreover, technology plays an important part in our lives, with numerous professions of interest using technology. Computers and smartphones have made enormous contributions to our everyday life in recent years, performing various computations and adjustments. The key advantage of such systems is the dependability with which authorized individuals may access the doors through a secure system with an interactive interface (not any one can access). In (Ghadekar et al., 2024), Athirah Nabihah M. et al. conducted research on the smart door lock, which incorporates Arduino-based and IoT-based systems and is a model for a door access system that enables users to observe the door system through transferrable Wi-Fi apps. This study seeks to be another way for locking and unlocking the door for individuals who forget to lock the door or who permanently lose the key. The common populace is constantly in a rush or is generally careless. This might explain why they forgot to shut the door to their home or location. It has become difficult to lock and unlock the door from a private location and to ensure that the door is adequately protected if we prefer to combine the area elsewhere. The following hardware components were used: Arduino, Servo Motor, Magnetic Contact Switch, NodeMCU (ESP 8266), Jumper Wire, Micro USB cable, and USB Type-B cable. They employed a prototyping model in their study. The requirement analysis is the first step in the prototyping model. The system's requirements are specified in depth at this phase. Throughout the process, system users are interviewed to learn about their expectations from the system. Apart from that, a questionnaire was created for users to complete, and it was discovered that the majority of respondents believe that this study would reduce the danger to their home's security. Home security includes both hardware and personal security procedures. Doors, locks, alarm systems, lights, motion detectors, and security systems are examples of security hardware. It is installed on a property; personal security measures include things like ensuring certain doors are latched, alarms are triggered, and windows are closed.

However, recent technological advancements in internet speeds and coverage, smartphone capabilities, and the development of mobile application technology have led to the increased adoption of smart home devices among homeowners who are looking to simplify routine daily tasks in a manner that is also more secure. Before to the invention of smart devices, homeowners would lock their doors every time they left the house, and some could even lose their keys, prohibiting them from entering or securing their houses, resulting in a significant loss of time for the homeowner. If the homeowner chooses to replace their house locks with an SDL, the difficulties listed above will be solved, and they will have greater control over the locks when away from home thanks to the power of the internet. As best of our knowledge, the current SDL systems lock the user into their proprietary environment and are prohibitively costly such as the developed SDLs by the large companies Nest x Yale and SimpliSafe. In addition, the current SDLs have a few usability problems. The first issue is that most smart locks require the user to use just a smartphone to manage them, without providing an alternative method of control.

The second issue is that the SLD requires the use of a hub device, which functions as a link between the smart lock and the internet. This study introduces IoT-SDL which enable the user to interact with it in a variety of manners, while also giving more functionality than current options and being easily configurable so that the user can make whatever modifications they need. Consequently, the user will supply with options when dealing with the IoT-SDR rather than restricting them to just one approach, and to provide the user with features that enable our suggested device to distinguish itself from the existing alternatives available as follow: (1) enable the user to manage the smart lock remotely using a web browser in the situation that their smartphone is not accessible, (2) enable the smart lock to operate independently of a hub device, (3) integrate two-factor authentication on hardware and software, (4) set up custom limits on the smart lock, and (5) create a feature that keeps track of who has accessed the device.

3. METHODOLOGY: CONCEPTUAL MODEL AND FUNCTIONS DESCRIPTION

In this section, the physical IoT-SDL system and its related algorithms are presented, with a focus on the functionality that these devices provide from the user's viewpoint. The user will be responsible for putting the IoT-SDL system on their door and initializing it by utilizing the Raspberry Pi's WIFI access point to input the home's WIFI SSID and password in order to enable internet connectivity to the IoT-SDL system, as illustrated in Figure 1. In addition, the user will be responsible for providing internet access to the IoT-SDL system. When the installation is completed, the user may access IoT-SDL through a web interface to setup users and enable IoT-SDL-supplied functionalities. This is achievable when the installation is completed. If the user activates the two-factor authentication option, the IoT-SDL has a touchscreen keypad and an RFID scanner that may be used to open the door.

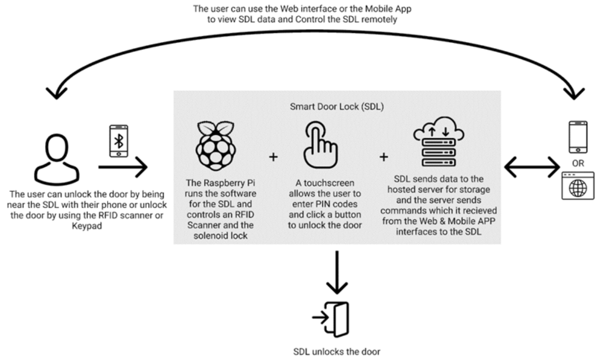

Figure 1. RC-SDL system architectural diagram

This will provide the user access to restricted locations that may be enabled by the user through the interfaces. Moreover, the user will be able to produce temporary PIN numbers that are valid for a certain period of time and are designed for dependable visitors to input on the IoT-SDL's touchscreen keypad in order to unlock the door. These codes will expire after a certain period of time and will no longer be useful. After using the interfaces that are given, the user will be able to remotely open the door with the push of a button. Using the various Application Programming Interfaces (APIs), the user will be able to access any and all of the data acquired by the IoT-SDL. The block diagram of the IoT-SDL system with the Raspberry Pi board is presented in Figure 2. The following components are connected to build the IoT-SDL system as shown in Figure 3: Raspberry Pi model B, RC522 RFID, GPIO Extension, Relay, Power supply, and the door lock mechanism. The Raspberry Pi is connected to the computer via USB cable in order to program the IoT-SDL system and power it up.

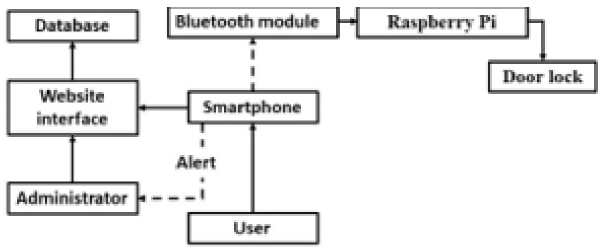

Figure 2. IoT-SDL block diagram

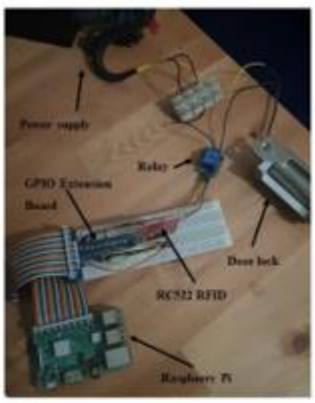

Figure 3. Hardware connection of the IoT-SDL system

Furthermore, there are two approaches to characterize IoT-SDL requirements: functional and non-functional. All of the prerequisites must be satisfied in any scenario.

Regarding the functional requirements, in granting access using Bluetooth, the IoT-SDL has the ability to identify an authorized user based on the smartphone they are using thanks to the Bluetooth low energy capabilities of the Raspberry Pi. Once identified, the door will immediately open for the approved user. By using APIs that are made available by the IoT-SDL, the Administrator will have the ability to add the Mac Address of the user's smartphone to a whitelist on the IoT-SDL. This will make it possible for the approval and access to be granted to the smartphone. In case of applying temporary PIN for granting access, the Administrator has the ability to provide temporary PIN codes to users through the interfaces, and the IoT-SDL will both convey these numbers to the user and store them. After then, the temporary PIN numbers may be entered into the keypad of the SDL in order to provide access to the temporary users. When RFID tags or cards are used as a way to allow access, the Administrator will be able to use the SDL's RFID scanner to program the RFID tags or cards, which will then be read by the SDL and used as a way to gain access. This will be possible whenever RFID tags or cards are used as a way to allow access. When RFID tags or cards are utilized for the purpose of giving access, this will become feasible. By the use of granting access interface tool, the Administrator is provided the ability to sign into any one of the IoT-SDL's interfaces in order to unlock the door remotely by clicking a button on the interface itself. This action causes a UNLOCK command to be sent to the IoT-SDL via the internet. In display access records through interfaces, The IoT-SDL will keep a record of each and every occurrence of access, and those data will be both saved and shown on the various interfaces. On the other hand, the quality characteristics of our IoT-SDL system are characterized by the non-functional criteria that are listed as performance, accessibility, and useability. In terms of performance, when an authorized user is at a distance considered acceptable for the device, the system should be able to recognize the user's smartphone in less than ten seconds. This should be the case when the user is at the distance that is considered suitable to the device. Considering the accessibility, in about a minute, the system should be able to capture and upload information on a user's successful or failed attempt to access the interfaces. The user interfaces for the systems be divided into three different categories in order to offer a user-friendly interface for the systems for useability. These categories are the home page, the data page, and the configuration page. The process starts with the Administrator establishing a connection between the device being used and a WIFI access point so that the system can be initialized

This allows the device to communicate with the access point. The transfer of data between the device and the access point is made possible as a result of this. Consequently, the device is now in a position to start interacting with the interfaces (either the Web App or the Mobile App). This, in turn, enables the Administrator to make changes to the device's parameters and add users to the system. After that, Administrator should fasten the device to the door, and then, Administrator should add users to the internal storage of the device so that it can begin functioning in the appropriate manner. When the device has been installed and configured, the Administrator should have full control over the device through the interface. This should take place after the device has been installed. The control ought to incorporate the ability to add or remove users, generate temporary PINs, view access records, enable or disable features, add specific authentication methods for each user, configure notification settings, and unlock the door remotely from a remote location. Hence, Administrator is the owner of the system and is considered a user but with higher privileges allowing him to manage other users and change the device settings while user is a normal user that is authorized to unlock the door using one of the three unlock methods provided by the system.

4. SYSTEM DIAGRAMS

"Lucid.App" and "Draw.io" are the tools that are used in the design process of the IoT-SDL diagrams. "Lucid.App" is used for the Class and ER diagrams because it enables to work together on the same diagram while also provides all of the tools that are necessary for developing effective diagrams. "Lucid.App." was chosen for the implementation of the Sequence and use case because it makes creating sequence diagrams more straightforward as shown in figure 4.

4.1. Class Diagram

The class diagram includes the SDL class in addition to the database class, the GUI class, and the APP class. Singleton class SDL is what interacts with the sensors and accepts input from the user to check for authentication in order to unlock the door. It also creates the access records that will be stored in the database, which is why it has references to database object. Since SDL is a singleton class, it also has references to database object. If the Two-Factor Authentication feature is turned on, the SDL class will also generate the OTP codes that will be sent to the user, and it will check to see if the user entered the correct OTP code before allowing them to unlock the door. The SDL class is the only one that uses the Database class because it is a singleton class. Records of the user object, access objects, and notification list are all kept in the database, which is a requirement for the functionality of the SDL system. Additionally, the user object itself is a requirement. It is the responsibility of the GUI class to create the graphical user interface (GUI) that will be displayed on a touchscreen to enable the user to input their credentials for authentication. This class makes use of the SDL module in order to perform manipulations on the hardware based on the results of authentication. It is the responsibility of the App class to dynamically alter the HTML code that is used for the pages in the webapp. This will make it possible to dynamically display information such as access records or user registered. The application class also makes use of the Flask library to respond to GET and POST HTTP requests. This gives us the ability to add users to the database by making use of the information that the user enters into the webapp forms that are implemented in the HTML code.

4.2. Sequence Diagram

Two different sequence diagrams that elaborate on how each type of user (Admin and User) will interact with the SDL system and how the SDL system will respond to the interaction have been designed. Figure 4 illustrates how a user will interact with the SDL device as well as how the SDL will handle the interaction in response to the user's actions. The user can initiate the interaction in one of three ways, which are to either approach the device with their registered mobile phone for Bluetooth Presence Detection, manually input data into the SDL using either the RFID scanner or the touchscreen keypad for PIN code entry, or they can approach the device without their registered mobile phone but still use Bluetooth Presence Detection. After this, the system will use the authentication method type and input value to search the database for a user object that is a match to the values entered; if a match is found, the database will return a reference to the user's object, which the SDL will use to send an OTP code to the user's email or mobile number. If a match is not found, the database will not return a reference to the user's object. The SDL will wait for the user to enter the OTP code before verifying it. If the code is valid, the SDL will generate an access object, save it in the database, and then unlock the door. If the one-time password (OTP) that the user enters is incorrect, the SDL will generate an access object, save it in the database, and display an error message on the screen. This will inform the user that the authentication attempt was unsuccessful, and they will need to try again by going through the entire process once more.

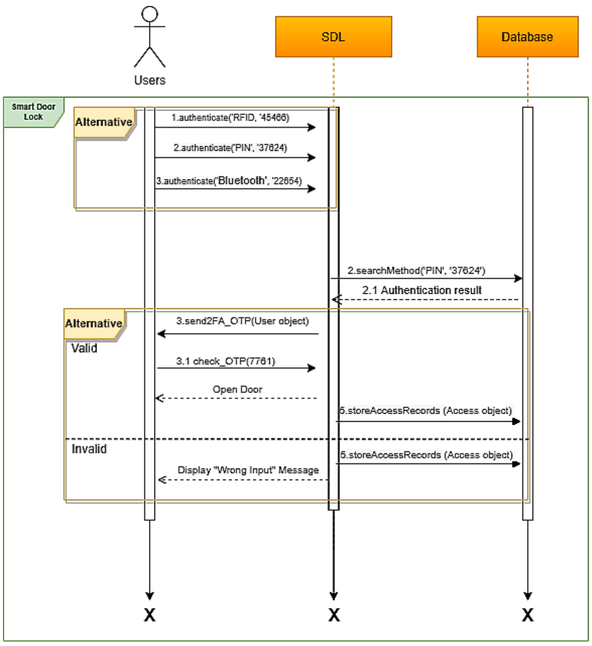

Figure 4. SDL User Sequence Diagram

Figure 5 shows how an Administrator will interact with the SDL device and how the SDL will handle the interaction. The Administrator can perform 5 actions with the SDL system such as Configure the system by passing a config object initially and then modifying that object using the reference to it, adding/removing users by passing the user object to the SDL then the SDL will interact with the database to do the task, remotely unlocking the door using the one of the interface (website or mobile app) which will send a command to the SDL to unlock the door and then create an access object to store in the database, and the last action that can be performed by the Administrator is to request all access record of a specific user by passing the user object to the SDL which will search the database for all access object that reference that same user object passed by the Administrator

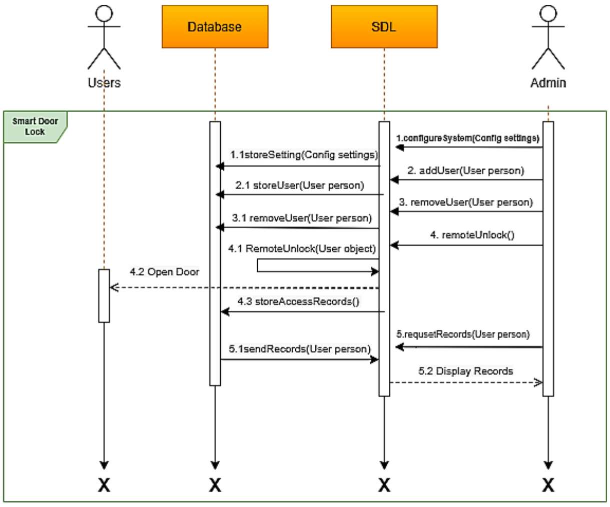

Figure 5. SDL Admin Sequence Diagram

4.3. ER Diagram

The relationship between the multiple entities in the SDL system. Our ER diagram has 4 entities. The SDL entity uses the config object in a 1 to 1 cardinality, uses the database object in a 1 to 1 cardinality, creates access objects in a 1 to 0 or more cardinality, authenticates user objects in 1 to 0 or more cardinality. The SDL entity's primary key is the owner attribute. The Database entity is used by the SDL entity to store and retrieve data in a 1 to 1 cardinality. The GUI entity uses the database to authenticate users and to record their access attempts whether they are successful or not. The GUI entity also uses the SDL entity to gain access to the hardware components in certain processes such as when the GUI entity needs to scan RFID tags, scan nearby Bluetooth devices, or open the door. The app entity uses the database to fetch and insert records. It fetches records to fill the tables used in the HTML code to display the data for the tables. It inserts records when users are being added to the database using the webapp forms. The app entity also uses the SDL entity to scan nearby Bluetooth devices and to unlock the smart door lock.

5. SYSTEM CODE AND TESTS

In our implementation we decided to have separate codes for each interface. The app.py file is the python code which uses the Flask library and is responsible for the web interface. Flask allows us to invoke python code based on HTTP requests made to the webserver and it manipulates the HTML code for the webpages using the templating language Jinja which allows for a dynamic display of information. Jinja achieves this by allowing us to include pseudo-python code in the HTML files where certain python variables will be displayed. The GUI.py file is the python code that uses the Tkinter library and is responsible for creating and displaying the GUI that is displayed on a touchscreen connected to the Raspberry Pi, and it allows the user to input their credentials for authentication. Both app.py and GUI.py files are supposed to be run immediately on boot for the system to function. Other than the app.py and GUI.py we also have two other files named SDL.py and DBconn.py these files act as modules providing useful functionality used by the interfaces without making them redundant. The GUI.py file starts by initializing the GUI window which will contain all the buttons. The buttons can either input data or process data. Buttons have a command attribute, and this dictates what action is performed by the button when it is clicked. We decided to make an event handler method which performs different functions depending on the value passed to it. When the button is used for input then the value passed to the event handler method is a positive number, and when the button is used to process data then the value passed to the event handler method is a negative number as shown in figure 6.

Figure 6. GUI result

The app.py file uses Flask to provide specific HTML files based on the URL requested by the browser, and this is achieved by declaring all possible URL routes in the python code and giving each of them a unique method to be performed.

The URL for this method declared as "/Home.html" and it can accept both GET and POST http requests. The method first displays data for today's usage of the smart door lock which it fetches from the database using today's date this returns a tuple of tuples from the database which includes all the records matching the SQL query. The data received from the query needs to be formatted before being used in the HTML code that is why we passed it to the Usage method. This same process is repeated below depend on the data received in the HTTP requests. The finalized data that has been formatted is then passed to the render_template method.

The HTML template will dynamically display data from the database. we are looping over data variable which we passed to the render_template method and then adding the information from it into the table data html tags. For each web page we have a similar process is used to either display data or receive data from HTTP requests that will later be used to preform specific functions. An example of this is the remote unlock method used on our webpage which unlocks the smart door lock if it receives a HTTP POST request which contains the variable remote as TRUE. The DBconn.py file is a file that provides a connection to the database. The reason we made it into a separate file is to make it into a singleton connection meaning only one instance of the connection is shared by all other files in the smart door lock. Since this file is considered a module when it is imported by any other code by using the same import syntax then the python interpreter will not make a new instance of this connection and instead use the instance stored in sys.modules this means it is a singleton and this implementation of a singleton is considered the most pythonic meaning it is the way of least astonishment.

The sdl.py file is a module that contains all the methods that directly use the hardware and the methods that are used by multiple files in the system. An example of methods that directly use the hardware is the open() method which sends a HIGH signal to the relay's GPIO pin in this case it is pin 12 which will open the door. Another example is the scanForBluetooth() method which is used by both interfaces and also directly uses the hardware to scan for nearby Bluetooth devices and returns them in the form of a list containing the name and mac address of the devices detected.

We utilized the PyTest Python package for unit testing, which enables us to use the python keyword assert to test functions with an expected value to ensure they perform as anticipated. We evaluated two methods, stars() and randomOTP(), to ensure they returned accurate results when called. The following tests were performed directly on the Raspberry Pi to ensure that they would work properly in the environment for which they were designed.

The integration testing was performed but running both the GUI.py and app.py at the same time with one database connection shared by them. All access recorded by the GUI.py code was immediately shown in the webapp, and all user credentials added using the webapp were immediately available to the GUI.py file to authenticate users meaning if a user was added using the webapp then he can immediately use his credentials to unlock the smart door lock using the GUI.py interface.

All of this was possible because of the singleton database connection meaning all changes made to the database are made available to both interface files as soon as they are committed by the code, and we made sure to commit all SQL queries invoked immediately after their execution to make the database connection available for other functions. This seamless data insertion and fetching proves compatibility between the GUI.py file and app.py file and showed great performance results as the changes were immediate. In terms of stress testing and load testing we could not simulate this since the GUI.py code can only authenticate one user at a time which is the way we intended and that is suitable for a home environment, also stress testing the webapp did not seem feasible since the webapp is only available on the local network meaning no outside traffic is expected. The structure of the project allows for all codes to run independently meaning that the GUI.py file and the app.py file can both preform their functionality without the other, and this was achieved by distributing the functionality such as using the sdl.py file to perform all the hardware required functions such as scanning RFID cards or scanning Bluetooth devices or unlocking the door by sending 12v to the electric door strike through the relay sensor controlled by the Raspberry Pi. The distribution of functionality helped in multiple ways such as avoidance of repeated code segments in multiple locations, allowing for easy alterations to the code, and detecting source of errors easily.

6. RESULT AND DISCUSSION

The IoT Smart Door Lock (IoT-SDL) system incorporates two distinct user interfaces designed to provide information and authenticate users. The graphical user interface (GUI) enables authentication by verifying user credentials stored in the database, ensuring access is restricted to registered users without limitations. Concurrently, the web application interface offers additional functionalities such as access record viewing, temporary passcode generation, and remote unlocking capabilities.

The web application home page facilitates access management by displaying the same-day access records and enabling the creation of temporary passcodes for trusted individuals. These passcodes are set for specific durations using a date and time picker. The generated passcodes are verified for uniqueness against the database to prevent duplication.

The data page within the web application provides detailed access logs for user-selected dates and contact details of registered users while maintaining privacy. Temporary passcodes are documented with relevant metadata, such as the creator, recipient, registration date, and expiration date.

The settings page offers administrative functionalities, including managing temporary passcodes, adding new users, and assigning unique credentials. When a new user is registered, validation ensures that the username and passcode are unique within the database. The process involves generating random passcodes, with immediate feedback provided to indicate whether the creation was successful or not.

Users can also register their RFID tags and the MAC addresses of their devices for Bluetooth-based authentication. The system validates the uniqueness of these credentials, preventing conflicts or misuse. Feedback messages guide users throughout these operations, ensuring a seamless and secure user experience.

Despite the operational success of all input forms and authentication mechanisms, opportunities exist for enhancing the user interface and user experience. The prototype fulfills essential requirements, including enabling access for registered users via multiple authentication methods, displaying detailed access records, managing user accounts, and generating temporary passcodes.

7. CONCLUSIONS

Conventional doors have been used for generations with little variation in how they work or the usefulness they provide. The Developed IoT-SDL system are working to modify the way doors function in order to keep up with the technological advancements of the twenty-first century. All of the approaches and tools used in this study, including the system model, operating principle, functional and non-functional requirements, and software component, make doors smarter and more accessible. The basis for a smart door lock device is built, allowing users to get rid of all the hefty keys that they spend time searching for when they are lost or have difficulties duplicating when allowing others to access the door. This enabled user to remotely operate the device, allowing others to use the smart door lock, remotely unlock the door, and conveniently keep track of who used the smart door lock. The created IoT-SDL system is built using easily accessible electrical components, making it more economical and less expensive than existing market alternatives based on study of comparable devices. Both hardware and software are included in the price: The price of the hardware involves (Raspberry Pi 3 Model b, sensors, solenoid lock). The price of the software includes (web interface and server). The web interface of the product was basic and user-friendly. The designed IoT-SDL system is connected to the door, making it easy to install the controller unit; nevertheless, the safe installation of the solenoid lock is dependent on the design and material of the door. The limitation of the developed IoT-SDL system is that OTP codes is obtained using Twilio's Sandbox which currently only sends messages through WhatsApp and the client must initiate the conversation with the Sandbox. It is impossible to send OTP codes to mobile phone numbers directly due to local laws and regulations. Besides this, a PC power supply should be used to power the hardware (Raspberry Pi and the electric door strike) separately due to different voltages need by the components, but the only limitation is the size of the power supply not its functionality. As a future work, it is recommended to solve the above-mentioned limitation. Also, the feature of can be improved to be more advanced by allowing multiple SDLs to work together in a mesh network where multiple SDLs are controlled by one parent SDL that relays all information from the other SDLs to the interfaces.

In conclusion, the IoT-SDL system presents a robust foundation for a highly functional IoT-enabled smart door lock. With iterative improvements, it has the potential to deliver an even more sophisticated and user-friendly device in the future.

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